THE GYSIN MAP OF A SECTION-TRIVIALIZED INVERTIBLE SHEAF COMMUTES WITH PROPER PUSHFORWARD WHEN D MAPS INTO ITSELF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Section setting up the bivariant action (relative-cycles chapter, Section section-action, canonical tag 0H65). cited at statement line 2569. Fixes the data (S, delta) and the operation alpha cap - the lemma lives in.
- Gysin homomorphism for a triple (L, s, i) (Chow Homology, Definition chow-definition-gysin-homomorphism, canonical tag 02T8). cited at statement lines 2572-2573. Defines the Gysin maps i^* used in the statement.
- Pullback pairs and the induced square (Chow Homology, Remark chow-remark-pullback-pairs, canonical tag 0B6Y). cited at statement line 2581. Defines the diagram D' -> X' -> X <- D and the induced maps i' and g.
- Gysin maps on the level of cycles (Chow Homology, Remark chow-remark-gysin-on-cycles, canonical tag 0B6Z). cited at proof lines 2588-2590. Makes the statement meaningful: all operations are defined on the level of cycles.
- Equal c1-invariants give equal divisor cycles (Chow Homology, Lemma chow-lemma-equal-c1-as-cycles, canonical tag 02ST). cited at proof lines 2599-2600. Identifies f_*[W' cap D']_k with [W cap D]_k in the case W' not contained in D'.

Source reference. Original section 0H65: sources/stacks-chow/tags.txt, source lines 21410–21410. Cited at statement line 2569: fixes the ambient data (S, delta) and the action.

Source reference. Original definition 02T8: sources/stacks-chow/tags.txt, source lines 3687–3687. Cited at statement lines 2572-2573: defines the Gysin maps i^* in the statement.

Source reference. Original remark 0B6Y: sources/stacks-chow/tags.txt, source lines 14086–14086. Cited at statement line 2581: defines the diagram D' -> X' -> X <- D.

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Source reference. Original remark 0B6Z: sources/stacks-chow/tags.txt, source lines 14087–14087. Cited at proof lines 2588-2590: makes the statement meaningful on cycles.

Source reference. Original lemma 02ST: sources/stacks-chow/tags.txt, source lines 3672–3672. Cited at proof lines 2599-2600: f_* of the first divisor cycle equals the second.

Lemma 0.2. *Let (S, δ) be as in Section 0H65. Let $f : X' \to X$ be a proper morphism of schemes locally of finite type over S. Let $(\mathcal{L}, s, i : D \to X)$ be as in Chow Homology, Definition 02T8. Form the diagram*

$$\begin{array}{ccc} D' & \xrightarrow[i']{} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ D & \xrightarrow{i} & X \end{array}$$

as in Chow Homology, Remark 0B6Y. If $\mathcal{L}|_D \cong \mathcal{O}_D$, then $i^* f_* \alpha' = g_*(i')^* \alpha'$ in $Z_k(D)$ for any $\alpha' \in Z_{k+1}(X')$.

Proof. The statement makes sense as all operations are defined on the level of cycles, see Chow Homology, Remark 0B6Z for the Gysin maps. Suppose $\alpha = [W']$ for some integral closed subscheme $W' \subset X'$. Let $W = f(W') \subset X$. In case $W' \not\subset D'$, then $W \not\subset D$ and we see that

$$[W' \cap D']_k = \mathrm{div}_{\mathcal{L}'|_{W'}}(s'|_{W'}) \quad \text{and} \quad [W \cap D]_k = \mathrm{div}_{\mathcal{L}|_W}(s|_W)$$

and hence f_* of the first cycle equals the second cycle by Chow Homology, Lemma 02ST. Hence the equality holds as cycles. In case $W' \subset D'$, then $W \subset D$ and both sides are zero by construction. □